

The empirical recurrence for OEIS A234906: recovering a conjecture that is not written down

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Abstract

OEIS A234906 records “Empirical recurrence of order 87”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 392$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 87, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 87 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234906 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having its diagonal sum differing from its antidiagonal sum by 1, with no adjacent elements equal (constant-stress tilted 1×1 tilings)”. It has offset 1 and begins

1720, 12336, 88092, 692608, 5287952, 42777680, 333040552, 2723879976, ...

The entry states:

Empirical recurrence of order 87 (see link above).

As of the “Last modified” line on the live entry (Jun 20 2022 18:04:38, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 392$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 392$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 784$ exact terms of the model returns order 87 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{87} c_i a(n-i),$$

c_1	7	c_{23}	−336141011955128414	c_{45}	642050281953665126831898	c_{67}	−197914
c_2	261	c_{24}	−246071460583862521	c_{46}	−321976738467584647344613	c_{68}	1650692
c_3	−1814	c_{25}	2685661331357162251	c_{47}	−1003959935051593129286549	c_{69}	513073
c_4	−30369	c_{26}	1390121968183480279	c_{48}	551859901029028535243554	c_{70}	−43067
c_5	210067	c_{27}	−18202130767725157198	c_{49}	1360472084220924366830979	c_{71}	−10764
c_6	2121619	c_{28}	−6139498065668638593	c_{50}	−807134706024177989475770	c_{72}	899893
c_7	−14665992	c_{29}	105238685402909544673	c_{51}	−1594904078578862872922763	c_{73}	179657
c_8	−100648782	c_{30}	19379856470662523243	c_{52}	1009375349507290980649795	c_{74}	−14757
c_9	699032577	c_{31}	−521465149205749349838	c_{53}	1613688120190016109607653	c_{75}	−2329
c_{10}	3467855476	c_{32}	−27609069698329638165	c_{54}	−1079518193205023406288263	c_{76}	18497
c_{11}	−24357181795	c_{33}	2222906765290127911777	c_{55}	−1404810759378303158400164	c_{77}	22667
c_{12}	−90495561554	c_{34}	−134124833862008566447	c_{56}	986104993789271938923300	c_{78}	−1709
c_{13}	647829096583	c_{35}	−8176939884243641790777	c_{57}	1048296563328772464158644	c_{79}	−156
c_{14}	1839518430321	c_{36}	1298755773902865514032	c_{58}	−767389864442916714270760	c_{80}	1106
c_{15}	−13548052639410	c_{37}	26017760205883151526478	c_{59}	−667462023984785233218750	c_{81}	693
c_{16}	−29700380965551	c_{38}	−6374903514370456286256	c_{60}	506818926456500229514616	c_{82}	−46
c_{17}	227633178524390	c_{39}	−71733785916773751456420	c_{61}	360635964808321085320658	c_{83}	−15
c_{18}	386077366106136	c_{40}	23019493965480650498405	c_{62}	−282632002168229584388524	c_{84}	110
c_{19}	−3123374526160540	c_{41}	171577502260942636531896	c_{63}	−164286646455028175224319	c_{85}	4
c_{20}	−4076040912608142	c_{42}	−66587959108889192085100	c_{64}	132221220108464646820660	c_{86}	−1
c_{21}	35446867714119675	c_{43}	−356236304942192325049975	c_{65}	62620028093994619473345	c_{87}	2
c_{22}	35095600156878375	c_{44}	159565735589536805777474	c_{66}	−51472049836717043115254		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 87, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $87 + 4$ terms removed returns order 87 as well, so $\deg q_{\min} = 87 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $87 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 87 that this sequence satisfies — and that family turns out to have exactly one member.

References

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